

RENANCE

FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE

BIO 101 — Past Questions & Materials

by Renance

Course: BIO 101

Institution: Federal University of Technology, Akure

Sources in this pack: 7

Includes past questions and lecture notes

This study pack was compiled and restructured by Renance for BIO 101 students at Federal University of Technology, Akure. It brings together 7 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app

Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 7 — Cyssa hub100LVL materials1st SemesterBIO 1012025 BIO 101 PRACTICE QUESTIONS.pdf

Lecture/Notes · Google Drive harvest

**2024/2025 ACADEMIC SESSION
LEAD CITY UNIVERSITY
DEPARTMENT OF BIOLOGICAL SCIENCES**

BIO 101: GENERAL BIOLOGY I PRACTICE QUESTIONS

1. The standard and smallest unit of life is
 - A. cell
 - B. Zygote
 - C. Embryo
 - D. Unicellular
2. Organisms that are single-celled and maintain all the processes of life are
 - A. single cell
 - B. Complex Cells
 - C. Variable Cells
 - D. Unicellular cells
3. _____ developed the first compound microscope
 - A. Mathias Schleiden
 - B. Edward Jenner
 - C. Robert Hooke
 - D. Jassen
4. _____ is the matrix within the cell into which all the substances are dissolved into
 - A. Cytoplasm
 - B. Protoplasm
 - C. Ectoplasm
 - D. Endoplasm
5. _____ organelle is for breaking down complex substances within the cell
 - A. Ribosomes
 - B. Lysosomes
 - C. Gogli Complexes
 - D. Mitochondria
6. Galileo Designed a _____
 - A. Microscope
 - B. Telescope
 - C. Electron Microscope
 - D. Light Microscope

7. Van Leuwenhoeck designed the _____
 - A. Microscope
 - B. Telescope
 - C. Electron Microscope
 - D. Light Microscope
8. The energy in the cells is spent in the form of
 - A. Mitochondria
 - B. ATP
 - C. ADP
 - D. AMP
9. Smooth Endoplasmic reticulum does not contain ____
 - A. Ribosomes
 - B. Lysosomes
 - C. Gogli Complexes
 - D. Mitochondria
10. Projections and membranes within the chloroplast where reactions take place is called
 - A. Cisterna
 - B. Ribosomes
 - C. Lysosomes
 - D. ATP
11. _____ is the organ of locomotion in some cells
 - A. Flagella
 - B. Mitochondria
 - C. Ribosomes
 - D. Lysosomes
12. Photosynthesis reactions within the cell take place in the _____
 - A. Ribosomes
 - B. Lysosomes
 - C. Chloroplast
 - D. Mitochondria
13. Mathias Schleiden found out that plants' tissues are made up of _____
 - A. Ribosomes
 - B. Lysosomes
 - C. Cells

- D. Mitochondria
14. Protoplasm is made up of cytoplasm and _____
- A. Cisterna
 - B. Ribosomes
 - C. Lysosomes
 - D. Nucleus
15. To view viruses, you have to make use of _____
- A. Microscope
 - B. Telescope
 - C. Light Microscope
 - D. Electron Microscope
16. Lysosomes contains
- A. ATP
 - B. Enzymes
 - C. Chloroplast
 - D. Pigments
17. The fluid within the chloroplast is called _____
- A. ATP
 - B. Enzymes
 - C. Chloroplast
 - D. Stroma
18. Fluid-filled spaces in cells are called _____
- A. Vacuoles
 - B. ATP
 - C. Chloroplast
 - D. Stroma
19. Animals' cells consume new food substances by _____
- A. Photosynthesis
 - B. Phagocytosis
 - C. Chlorophyll
 - D. Thylakoids
20. Examples of prokaryotes _____
- A. Plant cells

- B. Animal cells
C. Mahogany Cells
D. Bacteria cells
21. The Swedish naturalist, who began the struggle to classify all living things by proposing a two name is _____.
A. Carolus Linnaeus
B. Gregor Mendel
C. Matthias Schleiden
D. Charles Darwin
22. The system of biological classification used today was created by who?
A. Charles Darwin
B. Carolus Linnaeus
C. Stephen Jay
D. Robert Hooke
23. Linnaeus evolved a system of nomenclature called _____.
A. Vernacular
B. Monomial
C. Polynomial
D. Binomial
24. The smallest taxon of classification is _____.
A. Kingdom
B. Family
C. Variety
D. Species
25. The scientific name of an organism is made up of its _____ and _____.
A. Genus and species
B. Genus and family
C. Kingdom and family
D. Family and species
26. Which kingdom comprises organisms with many cells with cell walls and a nucleus, make their own food, and cannot move?
A. Fungi
B. Animalia
C. Protista

D. Plantae

27. A change in an organism's surroundings that causes it to react is called ____

- A. A stimulus
- B. A response
- C. Development
- D. Energy

28. Which phylum of animals is also called flatworms?

- A. Porifera
- B. Coelenterata
- C. Platyhelminthes
- D. Nematoda

29. Which sub-group in the plant kingdom produces flowers?

- A. Angiosperms
- B. Fungi
- C. Mosses
- D. Ferns

30. Vascular bundles are not found in

- A. Gymnosperms
- B. Pteridophytes
- C. Angiosperms
- D. Bryophytes

31. The transport of male gametes in bryophytes and pteridophytes occurs through

- A. Wind
- B. Water
- C. Birds
- D. Insects

32. Fucoxanthin is found in

- A. Red algae
- B. Brown algae
- C. Green algae
- D. Blue-green algae

33. The first land-inhabiting plants are

- A. Pteridophytes

- B. Bryophytes
- C. Gymnosperms
- D. Angiosperms

34. Which of the following is not a characteristic feature of bryophytes?

- A. Presence of archegonia
- B. Water is essential for fertilization
- C. An independent photosynthetic sporophyte
- D. Motile sperms

35. A moss differs from a fern in having

- A. Swimming Sperms
- B. Alternation of Generation
- C. Dependent Gametophyte
- D. Independent Gametophyte

36. Which of the following groups produces spores, but lacks vascular tissue?

- A. Fungi
- B. Pteridophytes
- C. Gymnosperms
- D. Angiosperms

37. The protonema of moss is

- A. Diploid
- B. Haploid
- C. Sporophyte
- D. Sporangium

38. Which of the following is an amphibian of the plant kingdom

- A. Red algae
- B. Fungi
- C. Moss
- D. Pines

39. Gymnosperms do not bear fruits because

- A. They are not pollinated
- B. They lack ovary
- C. They produce spores
- D. They do not have seeds

40. The group of plants having vascular tissue is called ____

- A. Bryophyta
- B. Pteridophyta
- C. Angiosperms
- D. Tracheophyta

41. Which group has seeds enclosed in fruits?

- A. Gymnospermae
- B. Angiospermae
- C. Phanerogamae
- D. Pteridophytes

42. Ferns belong to the following group

- A. Thallophyta
- B. Bryophyta
- C. Pteridophyta
- D. Gymnospermae

43. Seed-bearing plants include _____

- A. pteridophytes and gymnosperms
- B. bryophytes and pteridophytes
- C. gymnosperms and angiosperms
- D. bryophytes and angiosperms

44. What is the definition of a community?

- A. Two or more species of organisms in their environment
- B. The interaction between living organisms and their environment
- C. All the organisms of the same or closely related species
- D. Two or more populations of organisms

45. Association of animals when both partners are benefitted?

- A. Colony
- B. Mutualism
- C. Commensalism
- D. Amensalism

46. In a biotic community, the primary consumers are _____

- A. Detritivores
- B. Herbivores

- C. Carnivores
- D. Omnivores

47. What is a food chain?

- A. A long chain made of food
- B. Process of preparing food
- C. Food were locked by chain
- D. Linear sequence of energy transfer from one organism to another

48. A _____ makes its own energy and begins the food chain

- A. Consumer
- B. Herbivores
- C. Producer
- D. Decomposer

49. A type of consumer that eats primary consumer is called _____

- A. Prey
- B. Herbivore
- C. Tertiary consumer
- D. Secondary consumer

50. A type of consumer that eats only meat is called a _____

- A. Herbivore
- B. Carnivore
- C. Omnivore
- D. Brucivore

51. Of the following, which is the correct sequence for the flow of energy in a food chain

- A. Producers ---> consumers ---> decomposers
- B. Consumers ---> decomposers ---> producers
- C. Decomposers ---> producers ---> consumers
- D. Consumers ---> producers ---> decomposers

52. Which of the following conditions is true about the food chain?

- A. Provide more energy for the next trophic level
- B. Provides food for succeeding organisms
- C. Consume energy from the next trophic level
- D. Do not pass energy to the next trophic level

Paper 2 of 7 — Cyssa hub100LVL materials1st SemesterBIO 101BIO 101 (Introduction To Plant Identification

Lecture/Notes · Google Drive harvest

BIO 101: GENERAL BIOLOGY I
PSB UNIT, AAU, AKUNGBA AKOKO
TOPICS: INTRODUCTION TO PLANT IDENTIFICATION
TAXONOMY AND SYSTEMATICS AND USE OF
TAXONOMIC KEYS.

INTRODUCTION

Renanced
06-02-2024

Plant taxonomy simply put is the study of orderly arrangement of plants.

Olorode and Illoh, 2012 states that plant taxonomy is the science of identification, classification and naming (nomenclature) of plants, implying that bio-systematic botany is the wider subject of taxonomy involving the gathering and assessment of data for classification purposes. Olorode and Illoh(2012) defined systematics as the scientific study of the kinds and diversity of organisms and of any and all relationships among them.

Stace (1980) defined plant taxonomy and plant systematics as that particular branch of botany dealing with plant variations, the investigation of the causes and the consequences of these variations.

Bio-systematics encompasses both plant and animal taxonomy.

We are all aware of the fact that to classify objects is a basic practical activity for every human being. In this regard, Hopwood (1959) says that the urge to classify object is a fundamental human instinct, like the pre-disposition to sin, it accompanies into the world at birth and stays with us to the end (death). In fact, we do witness some sort of classification in our daily life e.g. in a library when books are arranged according to specializations-Archeology, Anthropology, Biochemistry, Botany, Chemistry, Computer Science, Criminology e.t.c.

Heywood (1977) collaborated by Gill(1988) says that one of the major role of plant taxonomy is to produce a system of classification that best reflects the totality of the organisms similarities and differences. Plant taxonomy and biosystematics botany according to Jeffery (1982) is neither difficult nor complicated, many people thinks that it is such a learnt academic activity that cannot be readily understood, others regard it as old fashioned, a relic of by-gone age of science that no longer demands serious attention.

These are enormous number of living organism in the globe and they show considerable variations in their shapes and sizes. These are more than 500,000 species of plants in the world and many more are being discovered every year.

Arranging and naming of living organisms is the prime or basic of taxonomy and the need of such a system was felt as early as of the time of Aristotle (384 – 322BC).

The aims/objectives of plant taxonomy and bio systematic botany(plant systematic) are three (3) folds.

1. To provide a convenient method of identification, naming and classification
2. To provide a classification which as far as possible expresses the natural relationship of organisms
3. To detect evolution at work, discovering its processes and interpreting it's results.

There are three interrelated aspects of plant taxonomy and biosystematic botany.

- i. Identification
- ii. Description and nomenclature (Naming)
- iii. Classification

IDENTIFICATION

Renance
06-02-2024

It deal with the determination of a taxon that has been identified, similar to or different from an already known organism. Taxon can be defined as the category or group of organism(s) being described.

Plant identification can be achieved with the aid of taxonomic or botanical literature in form of illustrations, drawings or sketches, direct pictures (photographs), from pictorial books, from botanical or horticultural gardens or from the wilds, from Natural history museum(s) or by comparison with plants of known identities-pressed and labeled plant specimen such as herbarium specimens.

An herbarium is where pressed, dried and nicely (finely) preserved plant specimens usually in large collections are kept and arranged in a sequence of an accepted system of classification i.e Bentham and Hooker's British system. Hutchinson and Dalziel's system is also a British system, De-Jussiaeu- is a French system, Engler and Prantl's system – is a German system.

Descriptions and keys of biological textbooks, floras, herbals, manuals, periodicals, journals and other taxonomic publications are also used in plant identification.

2. DESCRIPTION AND NOMENCLATURE (NAMING)

Man has always been a nomenclaturist he has always used names for plants, animals and objects with or without special terminologies or system(s).

Nomenclature simply defines means a system of naming, this botanical nomenclature is system of naming plants

The allocation of names to Taxon- singular.Taxa-plural

Nomenclature deals with the determination or allocation of correct names to organisms.Thus,plant nomenclature deals with the determination or allocation of correct names to plant organisms.

CLASSIFICATION

This is the study of the theory and practical methods of organization of taxa. Classification involves grouping of organisms together on the basic of what they have in common or it involves grouping of organisms together on the basis of characters, features or traits that they have in common.

Taxonomic Character(s)

This is an attribute of an organism that can be considered to vary independently of any other attribute (Olorode, 2012).i.e the scenting white flowers and the specialized berries-hesperidium fruits of the Citrus plants of family Rutaceae.

Classification involve the placing of plants in categories or groups according to a particular plan or system i.e. Hutchinson and Dalziel British system.

TYPES OF TAXONOMY

These are two main(major) types of taxonomy

- (1). Pioneer phase – Alpha taxonomy, Classical taxonomy and (or) Descriptive taxonomy
- (2). Bio systematic phase – Omega taxonomy and (or) Experimental taxonomy

1. PIONEER PHASE – ALPHA TAXONOMY

Deuse
06-02-2024

This can be based on the morphology (external features, habit, form or structure) life span; economic importance or geographical distribution, e.t.c.

Morphology or habit can be herb, under-shrub, shrub, liana or tree life span habit may also be herbaceous – annual or it may be biennial or perennial e.g. herbaceous perennial as in banana\plantain plant or woody perennial as in Mango tree plant.

Plant(s) may be grouped based on economic importance as a medicinal plant or as cereal plant, fiber, (fibre) plants, legume plant, ornamental plant, vegetable plant, timber-wood plant e.t.c.

2. BIO-SYSTEMATIC PHASE-OMEGA TAXONOMY

Bio-systematic botany is a phase of botany that delimits plants into natural units experimentally and classifying or grouping them objectively as taxa of different magnitudes such as Kingdom, Division, Class, Order, Family, Tribe, Genus and species e.t.c.

The ultimate objective of bio-systematic botany is the arrangement of plants in such a manner that will give clues or ideas of their evolutionary (phylogenetic) sequence in the plant Kingdom.

Plant bio-systematics is also known as Experimental taxonomy or Omega taxonomy

There is another minor type of plant taxonomy known as folk taxonomy, folk botany, ethno-taxonomy or ethno-botany. Ethno-botany can be defined as the relationship between people, their culture and the plants in their environments.

In all cultures, there is a system of ordering plant and animal life, serving as a system of reference to particular organisms. Some of the systems of naming local plants are based mainly on the experience of the community with respect to utilization of the plants.

The people of Africa as in other continents of the world, Yoruba people inclusive have their local system of plant taxonomy that pre-dates today's introduction of formal scientific system of classification.

Among the Yoruba's of Nigeria, we find examples of purely utilitarian and artificial plant classifications.

Most common vegetables among the Yoruba can be referred to by common name "Efo" which will appropriately be a supra-generic category, thus all species of *Talinum* (Portulacaceae/Talinaceae) are "Efo Gbure" and all edible species of Amaranth leafy vegetable is botanically known as *Amaranthus caudatus*, *Amaranthus hybridus* or *A. viridis* or also spiny Amaranth (*Amaranthus*) known scientifically as *A. spinosus*.

Yorubas do mention Efos-Efo Ebolo, Ewuro, Odu, Worowo and Efo Yanrin, in this folk botany-taxonomy, Efo becomes the genus, while the epithets ebolo, ewuro, odu, worowo and yanrin are species.

Scientifically, Efo Ebolo is *Crassocephalum crepidioides* (Asteraceae)

Efo Ewuro-*Vernonia amygdalina* (Asteraceae)

Efo Odu-*Solanum nigrum* (Solanaceae)

Efo Worowo-*Crassocephalum bialafrae* synonym *Senecio bialafrae* (Asteraceae)

Efo Yanrin-*Lactuca taraxacifolia*(Asteraceae)

Also the *Citrus* genus(Rutaceae family) among the Yoruba's are Osans with specialized berry fruits-the hesperidium fruits-

Grape-Citrus *paradisi*.Lime-Citrus *aurantiifolia*.Lemon-Citrus *limoni*.Shaddock-Citrus *decumana*/Citrus *grandis*.Sweet orange-Citrus *sinensis* and Tangerine-Citrus *reticulata*. *Ogede*
06-02-2014

Yoruba's call Lime as Osan wewe/Orombo wewe and Sweet orange as Osan mimu/Orombo mimu.

Note that there are no local(vernacular nor native)names for grape,lemon,shaddock and tangerine,

The Yoruba's also call the highly divergent African Star apple as Osans-Osan Agbalumo-*Chrysophyllum albidum*(Sapotaceae family).

The herbaceous perennial-*Musa spp* with longish berry fruits-the edible banana-*Musa sapientum* L. is called Ogede were\jije by the Yoruba's and the edible boilable/cookable/fry-able plantain-*Musa paradisiaca* L. is called Ogede Agbagba and with Carolius(Karl)Linnaeus as the cited authority.

All these points to the inadequacies of the minor(local/native)folk taxonomy and the universality of alpha-taxonomy and omega-taxonomy(bio-systematics).

ECOLOGICAL(ECO)- TAXONOMY

Among the Yorubas, the corresponding generic name for *Terminalia* (Combretaceae) is Idi or Udi, thus are Idi Igbo or Udi Igbo and Idi odan or Udi odan (*Terminalia ivorensis* and *Terminalia* (T.) *glaucescens* respectively) Igbo and Odan are specific (species), ecologically derived epithets meaning forest and savannah environments respectively.

SYSTEMS OF CLASSIFICATION

There are two main (major) systems of classification

1. Artificial system
2. Natural system

ARTIFICIAL SYSTEM OF CLASSIFICATION

Artificial system of classification is the same thing or equal to the Alpha, Classical, Pioneer or largely descriptive, morphology (morphological) or external features; stage or level. As earlier mentioned, plants may be classified based on their habit, geographical location (habitat) or vegetation location, economic importance etc.

Attributes or characteristics of the artificial system of classification

The artificial system of classification is based on easily observable characteristics with emphasis on convenience and simplicity i.e.

- i. It contain very little scientific information
- ii. It has little predictive value
- iii. It is a special purpose classification chosen to suit the purpose from which it is designed

NATURAL SYSTEM OF CLASSIFICATION

The natural system of classification is the same thing or equal to the bio-systematic phase, experimental taxonomy or omega taxonomy stage.

A natural system of classification is based on overall similarities and differences in a number of attributes, characters, features or traits. The more the features a classification entails, the greater the information it carries or conveys.

The natural system of classification groups together organisms that are closely related phylogenetically (evolutionary/evolutionarily wise) and takes the genetic constitution of an organisms into consideration.

The attributes of a natural system of classification are: *Renanced*
06-02-2024

- i. Highly informational or highly informative scientifically.
- ii. Its has high predictive value
 - iv. It is based on overall similarities or over-all differences in a high number of traits. And as earlier said, the more the attributes, characters, features or trait a classification entails the greater the information its carries or conveys.

TAXONOMIC HIERARCHY/ TAXONOMIC HIERARCHIES(TAXONOMIC RANKS)

The arrangement of taxonomic categories in descending series from the highest to the lowest categories or vice versa is known as **Taxonomic hierarchy**.

The International Code of Botanical Nomenclature (ICBN) now named the International Code of Nomenclature for Algae, Fungi and Other Plants (ICNAFOP) which is the international agreed-rule book has recognized thirteen (13) major ranks and other minor ranks.

1. Kingdom

- Sub-kingdom
- 2. Division
 - Sub-Division
- 3. Class
 - Sub-Class
- 4. Order
 - Sub-Order
- 5. Family
 - Sub-Family
- 6. Tribe
 - Sub-Tribe
- 7. Genus
 - Sub-Genus
- 8. Section
 - Sub-Section
- 9. Series
 - Sub-Series
- 10. Species
 - Sub-Species
- 11. Variety
 - Sub-Variety
- 12. Form (Forma)
 - Sub-Form (Sub-Forma)
- 13. Clone
 - Sub-Clone – According to(After) - A.Clive. Stace (1980)

TYPES OF NOMENCLATURE

The two major (main) types of Nomenclature are

1. Polynomial Nomenclature
2. Trinomial Nomenclature
3. Binomial Nomenclature

Renanced
06-02-2024

Poly= Many=

Polynomial nomenclature involves naming with long descriptive :

Ranunculus folis ovatis serratis, scapo, nudo, uniflorus.

Meanings

Ranunculus with ovate, and serrated leaf with solitary, unisexual flowers- (one-sexed flowers) with naked scape:- A plant that is leafless, while flowering.

2) Trinomial Nomenclature- These are three(-3-) named nomenclature plant/(or) animal i.e *Hibiscus tiliaceus variegatus*

Homo sapiens sapiens

Cabbage: *Brassica oleracea capitata*

Binomial Nomenclature

This is a biological naming based on the binomial nomenclature bi = two nominal nomenclature – two named nomenclature pioneered by Carl (Carlus) (Karl) Linnaeus (1707-1778) in 1753 and lesserly Caspar/Gapar /Kaspar Bauhin (1560 -1624).

1753-was the starting point of the introduction of Binominal Nomenclature by him, in this binominal system, each organism either plant or animal organism must have two (-2-) Latinized names.

A Generic (Genus) Name beginning with a capital letter and a species (specific)name=specific epithet beginning with a lower case letter, for example the new binominal of Ranunculus and bulbosus is more convenient instead of the former polynomial of Ranunculus folis ovatis scape nudo uniflorus.

BIO 101:GENERAL BIOLOGY 1.

PLANT SCIENCE AND BIOTECHNOLOGY UNIT

ADEKUNLE AJASIN UNIVERSITY,AKUNGBA AKOKO

PART TWO(2):USE OF TAXONOMIC KEYS

Specimen Identification and the Use of Taxonomic (Identification) Keys.

According to the English Dictionary,a key is a word or a list of symbols for interpreting a code or a set of answers to problems or something that provides access, or control or insight.

A Taxonomic key is an outline prepared for use in identifying both plant and animal organisms by a dual process of either confirmation or elimination.

In biology; an identification key or Taxonomic key is usually a printed or more recently, rarely a computer aided device that aids the identification of biological entities such as plants, animals, fossils, micro-organisms, pollen grains and spores.

Identification keys are also used in other scientific and technical fields to identify diseases, soil types, geological minerals and archeological materials.

Recently there has been considerable research in developing computerized keys that have multiple entries derived from a detailed data base of the taxa. The inter-active nature of computerized keys makes them highly desirable and allow users to identify a plant. As software programs improve there is the likelihood of increase use of computer generated keys.

Renanced
06-02-2024

The main purpose of an identification key or taxonomic key is to facilitate identification or to distinguish one type or one kind of organism from another.

Identification errors may have serious consequences in both pure and applied disciplines

Pure discipline i.e ecology

Applied discipline i.e medical diagnosis, forensic science etc. therefore identification keys must be constructed with great care in order to minimize the incidence of such errors.

A Taxonomic key reflect ideas of evolutionary or phylogenetic relationships if it is a natural key and vice versa if it's at an artificial key, it will not reflect information of evolutionary or phylogenetic relationships.

Before one can key vascular plants for example, it is necessary to gain an understanding of basic morphological terms.

Reproductive characters- those pertaining to the asexual sporangia, to the sexual cones, flowers, seeds or fruits are commonly used in keys because they are more reliable since they usually show little, if any variation. When characters are stable and not environmentally variable, they are termed conservative.

In contrast, vegetative characters, e.g. leaf are more variable, more so, when individuals of the same plant species are grown in differential environmental regimes/conditions.

This explains why keys using only vegetative character such as leaf is not as reliable as keys using both vegetative and reproductive characters. It is highly

recommended that both vegetative and reproductive characters be included in a Taxonomic key or Identification key.

Keys provide statements called leads numbered 1, 2, 3 and so on, each lead dealing with a particular observable character of the plant.

Character(s) used in Taxonomic keys are usually observable morphological features and may be qualitative such as leaf, stem, roots, flower, seed and fruits or quantitative such as number(s) of leaves, stem, root, flowers, seed and fruits.

Both qualitative and quantitative characters may be used, but the character/characteristic must be (a) constant for a species and not subject to variation as a result of environmental influences. In this respect, size is often a bad example, since size can be readily influenced by the environment [Environmental factors].

Types of keys

There are various types of diagnostic keys in taxonomic studies, but it can be divided into two major (main) key types

1. Natural Key and
2. Artificial Key

1. Natural Key

In Natural Key, many character(s), characteristics are observable scientifically research wise on the plant for example – a) the Anatomy- Gross internal structure - Anato-Taxonomy, b) the Chemistry/Phyto-Chemical or chemical constitution- Chemo- Taxonomy c) [Chromosome constitution- Cyto-Taxonomy and the Palynology-Pollen Grain Types, Nature of Aperture and Nature of spores- Palyno-Taxonomy.

Natural key is at the level of Bio-systematic botany experimental Taxonomy or Omega –Taxonomy equals to the Natural system of classification.

2. Artificial key

a. Numerical/Arithmetical Key is an example of an Artificial Key

b Dichotomous Key-not within the scope of this introductory class level

2a. Arithmetical/Numerical Key is the use of arithmetic or numerical codes in plant identification and classification - e.g. for Leaf

Arrangement

For example, if the character or these characteristics are present, such character/characteristics will be indicated by writing one (1) if such character/character are not present, it will be indicated by writing down zero (0).-an issue based on either 0 or 1.

i.e.

Compound = 1

or

Simple = 2

Pinnate = 3

Obovate = 4

Palmately lobed = 5

Parallel Veined (Venation) = 6

Net (Reticulate) Veined (Venation) = 7

Gmelina tree-Gmelina arborea tree leaf: =0100001

Pawpaw tree- Carica papaya Linn Leaf = 0100101

Siamea Tree- Cassia siamea/ Senna siamea Leaf = 1010001

Country almond/Indian almond Terminalia catappa tree Leaf = 0001001

Another example of the character or characteristics are present, in the case; such character/characteristics will be indicated by verifying the actual (real) number/ numeral involved, but if such character/ characteristics are not present, it will be indicated by writing down/ zero (0).

Compound = 1

Simple = 2

Pinnate = 3

Obovate = 4

Palmately lobed = 5

Parallel veined (Venation) = 6

Net (Reticulate) Veined (Venation) = 7

Renanced
06-02-2024

Gmelina arborea = 0200007

Pawpaw- Carica papaya leaf = 0200507

Siamea Tree – Cassia siamea/ Senna siamea leaf = 1030007

Cassia is a synonym of Senna, that is Cassia is synonymous to Senna Genus. Synonym refers to the discarded name or to the series of discarded names applied to a single taxon.

Country almond/Indian almond – Terminalia catappa leaf = 0204007

Artificial key is at the level of alpha, classical, pioneer or largely descriptive, morphology (morphological) or external features stage or level; equals to the artificial system of classification.

Advice for using taxonomic key/identification key

1. Never guess at all- if you don't know the precise meaning of terms used in plant descriptions; consult the list of abbreviations and botanical glossary or botanical dictionary for meaning of terminologies.
2. If the plant features are very small, use appropriate magnifying lenses to inspect them clearly and carefully.
3. Where measurements are required, use a graduated ruler, never guess sizes. Give exact measurements or ratio's. The terms large and small have no meaning in the absence of quantitative records.

Paper 3 of 7 — Cyssa hub100LVL materials1st SemesterBIO 101BIO 101- AEB Combined notes-24-25.pdf

Lecture/Notes · Google Drive harvest

Topic 1

Cell Structure & Organization

Cells are the basic, functional unit of life. The smallest independent component that we would find if we break apart an organism to cellular level is the cell,

A cell is defined as the smallest, basic unit of life that is responsible for all of life's processes.

Types Of Cells

- Prokaryotic cells: They have no nucleus. Instead, some prokaryotes such as bacteria have a region within the cell where the genetic material is freely suspended. This region is called **Nucleoid**. They are single-celled micro-organisms. Examples include archaea, bacteria and cyanobacteria.
- Eukaryotic cells: They have a true nucleus. They involve plants, fungi, protozoans and animals.

Cell Structure

The cell structure comprises individual components with specific functions essential to carry out life's processes. These components include: Cell wall, Cell membrane, and Cytoplasm, Nucleus and Cell organelles.

Organization of the Cell

1. Cell: cells are defined as the building blocks and basic units of life in living organisms. They are responsible for regulating the biological processes in organisms.
2. Tissue: cells, when grouped together to perform a function, are known as tissues. Both plants and animals have tissues. Examples of animal tissue include connective, nervous, epithelial and muscle tissues while an example of plant tissues are epidermal, vascular and ground tissues.
3. Organ: these are defined as the group of tissues that are responsible for performing specific functions. Organs such as kidneys, heart, lungs and others are the characteristic structures that perform several functions.
4. Organ system: Different organs combine together to form organ systems: examples of organ systems are the circulatory system, digestive system and respiratory system.

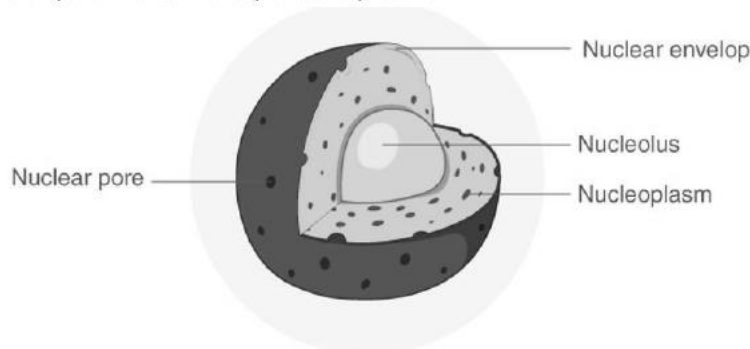
Topic 2

Function of Cellular Organelles in Animal cells

The Nucleus

The nucleus is the largest organelle in a eukaryotic cell and is considered to be the cell's control center. It contains most of the cell's DNA, which makes up chromosomes and is encoded with the genetic instructions for making proteins. The function of the nucleus is to regulate gene expression, including controlling which proteins the cell makes. In addition to DNA, the nucleus contains a thick liquid called nucleoplasm that is similar in composition to the cytosol found in the cytoplasm outside the nucleus. Most eukaryotic cells contain just a single nucleus, but some types of cells, such as red blood cells, contain no nucleus. A few other types of cells, such as muscle cells, contain multiple nuclei.

The nucleus is surrounded by a structure called the nuclear envelope, which contains tiny perforations, or pores. The nucleus also contains a dense center called the nucleolus. The membrane enclosing the nucleus is called the nuclear envelope. This is actually a double membrane that encloses the entire organelle and isolates its contents from the cellular cytoplasm. Tiny holes, called nuclear pores, allow large molecules to pass through the nuclear envelope with the help of special proteins. Large proteins and RNA molecules must be able to pass through the nuclear envelope so proteins can be synthesized in the cytoplasm and the genetic material can be maintained inside the nucleus. The nucleolus is mainly involved in the assembly of ribosomes. After being produced in the nucleolus, ribosomes are exported to the cytoplasm where they are involved in the synthesis of proteins.



Mitochondria

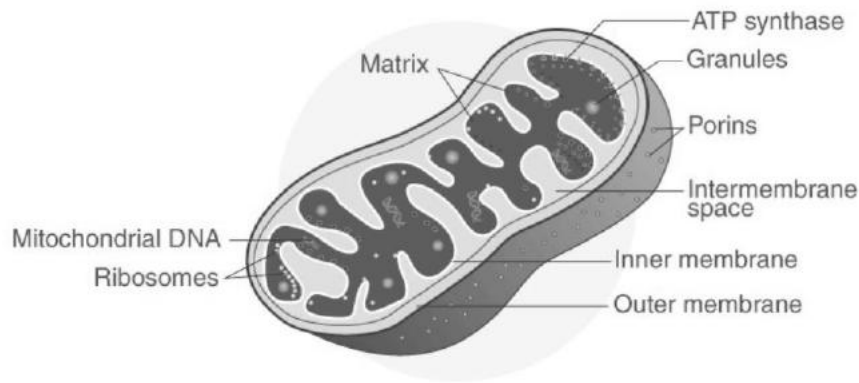
The mitochondrion (plural, mitochondria) is an organelle that makes energy available to the cell. This is why mitochondria are sometimes referred to as the power plants of the cell. They use energy from organic compounds such as glucose to make molecules of ATP (adenosine triphosphate), an energy-carrying molecule that is used almost universally inside cells for energy.

Mitochondria, organelles specialized to carry out aerobic respiration, contain an inner membrane folded into cristae, which form two separate compartments: the inner membrane space and the matrix. The Krebs Cycle takes place in the matrix. The electron transport chain is embedded in the inner membrane and uses both compartments to make ATP by chemiosmosis. Mitochondria have their own DNA and ribosomes.

Mitochondrial Compartments

The double membrane nature of the mitochondria results in five distinct compartments, each with an important role in cellular respiration. These compartments are:

the outer mitochondrial membrane,
the intermembrane space (the space between the outer and inner membranes),
the inner mitochondrial membrane,
the cristae (formed by infoldings of the inner membrane), and
the matrix (space within the inner membrane).



Endoplasmic Reticulum

The endoplasmic reticulum (ER) (plural, reticuli) is a network of phospholipid membranes that form hollow tubes, flattened sheets, and round sacs. These flattened, hollow folds and sacs are called cisternae. The ER has two major functions:

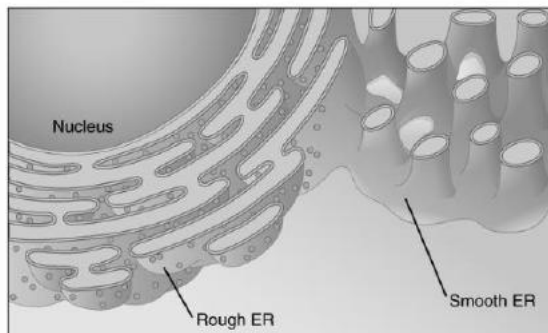
Transport: Molecules, such as proteins, can move from place to place inside the ER, much like on an intracellular highway.

Synthesis: Ribosomes that are attached to the ER, make proteins. Lipids are also produced in the ER.

There are two types of endoplasmic reticulum, rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER):

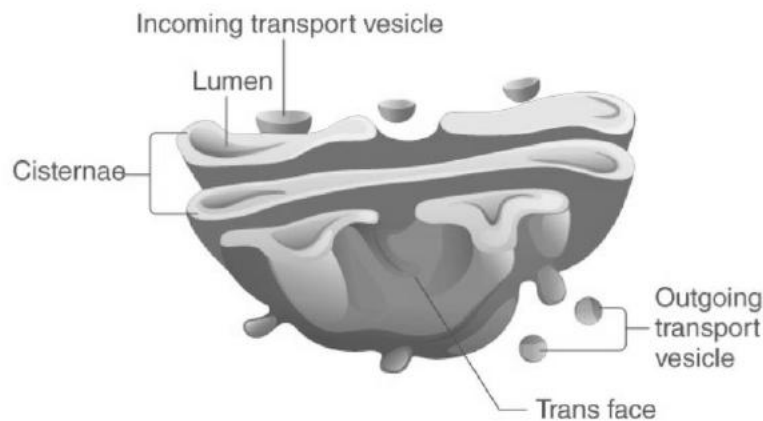
Rough endoplasmic reticulum is studded with ribosomes, which gives it a “rough” appearance. These ribosomes make proteins that are then transported from the ER in small sacs called transport vesicles. The transport vesicles pinch off the ends of the ER. The rough endoplasmic reticulum works with the Golgi apparatus to move new proteins to their proper destinations in the cell. The membrane of the RER is continuous with the outer layer of the nuclear envelope.

Smooth endoplasmic reticulum does not have any ribosomes attached to it, and so it has a smooth appearance. SER has many different functions, some of which include lipid synthesis, calcium ion storage, and drug detoxification. The smooth endoplasmic reticulum is found in both animal and plant cells and it serves different functions in each. The SER is made up of tubules and vesicles that branch out to form a network. In some cells, there are dilated areas like the sacs of RER. Smooth endoplasmic reticulum and RER form an interconnected network.



Golgi Apparatus

The Golgi apparatus is a large organelle that processes proteins and prepares them for use both inside and outside the cell. The Golgi apparatus modifies, sorts, and packages different substances for secretion out of the cell, or for use within the cell. The Golgi apparatus is found close to the nucleus of the cell where it modifies proteins that have been delivered in transport vesicles from the Rough Endoplasmic Reticulum. It is also involved in the transport of lipids around the cell. Pieces of the Golgi membrane pinch off to form vesicles that transport molecules around the cell. The Golgi apparatus can be thought of as similar to a post office; it packages and labels "items" and then sends them to different parts of the cell. The Golgi apparatus tends to be larger and more numerous in cells that synthesize and secrete large quantities of materials; for example, the plasma B cells and the antibody-secreting cells of the immune system have prominent Golgi complexes. The Golgi apparatus manipulates products from the Rough Endoplasmic Reticulum (ER) and also produces new organelles called lysosomes. Proteins and other products of the ER are sent to the Golgi apparatus, which organizes, modifies, packages, and tags them. Some of these products are transported to other areas of the cell and some are exported from the cell through exocytosis. Enzymatic proteins are packaged as new lysosomes.



The stack of cisternae has four functional regions: the cis-Golgi network, medial-Golgi, endo-Golgi, and trans-Golgi network. Vesicles from the ER fuse with the network and subsequently progress through the stack from the cis- to the trans-Golgi network, where they are packaged and sent to their destination. Each cisterna includes special Golgi enzymes which modify or help to modify proteins that travel through it.

Vesicles and Vacuoles

Both vesicles and vacuoles are sac-like organelles that store and transport materials in the cell. Vesicles are much smaller than vacuoles and have a variety of functions. The vesicles that pinch off from the membranes of the ER and Golgi apparatus store and transport protein and lipid molecules.. Other vesicles include:

Lysosomes, which use enzymes to break down foreign matter and dead cells.

Peroxisomes, which use oxygen to break down poisons.

Transport vesicles, transport contents between organelle as well as between cell exterior and interior.

Centrioles

Centrioles are organelles involved in cell division. The function of centrioles is to help organize the chromosomes before cell division occurs so that each daughter cell has the correct number of chromosomes after the cell divides. Centrioles are found only in animal cells and are located near the nucleus. Each centriole is made mainly of a protein named tubulin. The centriole is cylindrical in shape and consists of many microtubules, as shown in the model pictured below.

Ribosomes

Ribosomes are small structures where proteins are made. Although they are not enclosed within a membrane, they are frequently considered organelles. Each ribosome is formed of two subunits. Both subunits consist of proteins and RNA. RNA from the nucleus carries the genetic code, copied from DNA, which remains in the nucleus. At the ribosome, the genetic code in RNA is used to assemble and join together amino acids to make proteins. Ribosomes can be found alone or in groups within the cytoplasm as well as on the RER.

Review

Describe the structure and function of the nucleus.

Explain how the nucleus, ribosomes, rough endoplasmic reticulum, and Golgi apparatus work together to make and transport proteins.

Why are mitochondria referred to as the power plants of the cell?

What roles are played by vesicles and vacuoles?

Why do all cells need ribosomes?

Lysosomes and peroxisomes are types of:

- A. Organelles
- B. Vesicles
- C. Vacuoles
- D. Both A and B

Which of the following organelles fits best with each description of function? Choose only one organelle for each answer: Golgi apparatus, centrioles, nucleolus, nucleus, rough endoplasmic reticulum

- a. Contains the genetic instructions for the production of proteins
- b. Organizes chromosomes before cell division

c. Provides a framework for ribosomes

d. Packages and labels proteins

e. Assembles ribosomes

True or False. All eukaryotic cells have a nucleus.

True or False. The outer surface of the nucleus of a eukaryotic cell is not completely solid.

Topic 3

CHARACTERISTICS AND CLASSIFICATIONS OF LIVING THINGS

What are living things?

Living things are organisms that have the characteristics of life and are made up of cell(s). They are group of organisms that have a fixed life cycle. They could be microscopic or macroscopic and are capable of exhibiting movement, growth, reproduction, and response to stimuli.

Characteristics of Living Things

All living organisms share certain fundamental characteristics that distinguish them from non-living things. These characteristics include:

1 Cellular Organization

All living things are composed of one or more cells. Cells are the basic unit of life, performing all necessary biological functions. Organisms can be unicellular (e.g., bacteria) or multicellular (e.g., humans, plants).

2. Growth and Development

Living things grow by increasing in size and complexity but the growth occurs due to cell division and enlargement. Development involves changes in form and function over time (e.g., metamorphosis in butterflies).

3. Metabolism

Living things carry out biochemical reactions to obtain and use energy. It could be anabolism (the building up of complex molecules e.g., photosynthesis) or Catabolism (the breaking down of molecules to release energy e.g., respiration).

4. Reproduction

The ability to produce offspring ensures the survival of a species. Reproduction could be;

- ❖ Asexual reproduction: Offspring arise from a single parent (e.g., binary fission in bacteria).
- ❖ Sexual reproduction: Offspring result from the fusion of male and female gametes (e.g., humans, flowering plants).

5. Response to Stimuli (Irritability)

Living things detect and respond to environmental changes (stimuli). Examples include;

- Plants bending toward light (phototropism).
- Humans pulling their hands away from a hot surface (reflex action).

6. Homeostasis

This is the ability to maintain a stable internal environment despite external changes. Examples include regulation of body temperature in humans and maintaining water balance in plants and animals.

7. Movement

All living things exhibit some form of movement. Animals move using muscles, while plants exhibit growth movements (e.g., roots growing toward water).

8. Adaptation

Living organisms develop structural, behavioral, or physiological features that help them survive in their environment. Examples are Camouflage in chameleons and thick fur in polar bears for insulation.

9. Excretion

This is the process of removing metabolic waste from the body. There are three forms of metabolic waste in animal; Ammonia, Uric acid, and Urea. Humans excrete urea through urine. Carbondioxide and oxygen are also examples of metabolic waste. Plants excrete oxygen as a by-product of photosynthesis.

10. Nutrition

This is the process of obtaining and utilizing food for energy and growth. It could be;

- ❖ Autotrophic nutrition: Organisms make their own food (e.g., plants via photosynthesis).
- ❖ Heterotrophic nutrition: Organisms obtain food from other organisms (e.g., animals, fungi).

Understanding the characteristics of living things helps differentiate them from non-living matter.

CLASSIFICATION OF LIVING THINGS

Living organisms are classified based on shared characteristics to make their study easier. The science of classification is called taxonomy.

A. Levels of Classification

Living organisms are grouped into hierarchical categories:

1. Kingdom – Broadest classification (e.g., Animalia, Plantae).
2. Phylum – Groups organisms with similar body structures.
3. Class – Further division within a phylum.
4. Order – Groups organisms based on even more specific similarities.
5. Family – Closely related organisms.
6. Genus – Groups species with similar characteristics.
7. Species– The most specific level; organisms can interbreed to produce fertile offspring.

A typical example of the above level of classification

- ❖ Human Classification:
 - Kingdom: Animalia
 - Phylum: Chordata
 - Class: Mammalia
 - Order: Primates

- Family: Hominidae
- Genus: Homo
- Species: Homo sapiens

B. Five Kingdom Classification

The most widely used classification system groups organisms into five kingdoms:

1. Kingdom Monera (Prokaryotae)

This includes bacteria and archaea. They are Unicellular and lack a nucleus (prokaryotic). Examples are *Escherichia coli*, *Streptococcus spp.*

2. Kingdom Protista

They are mostly unicellular, with a nucleus (eukaryotic). Some are autotrophic (e.g., algae), others heterotrophic (e.g., protozoa). Examples are Amoeba, Paramecium, Euglena.

3. Kingdom Fungi

They are mostly multicellular (except yeast). They absorb nutrients from organic material (heterotrophic). Examples are Mushrooms, Yeasts, Molds.

4. Kingdom Plantae

They are multicellular, autotrophic (photosynthetic) and have cell walls made of cellulose. Examples include Mosses, Ferns, Flowering plants.

5. Kingdom Animalia

They are multicellular like Kingdom Plantae but **heterotrophic**. No cell walls and are highly mobile. Examples are Insects, Fish, Mammals.

Three-Domain System (Modern Classification)

In modern taxonomy, organisms are classified into three domains based on genetic and molecular differences:

1. Bacteria – True bacteria and unicellular prokaryotes.
2. Archaea – Prokaryotes that live in extreme environments.
3. Eukarya– Includes protists, fungi, plants, and animals (all eukaryotic organisms).

In summary, classification organizes biological diversity, making it easier to study and understand the relationships among organisms.

REPRODUCTION

Reproduction can be defined as a biological process (that is living organisms; not chemical process) in which an organism gives rise to young ones (offspring) similar to itself. The offspring grow, mature and in turn produce new offspring. Thus, there is a cycle of birth, growth and death. Reproduction enables the continuity of the species, generation after generation.

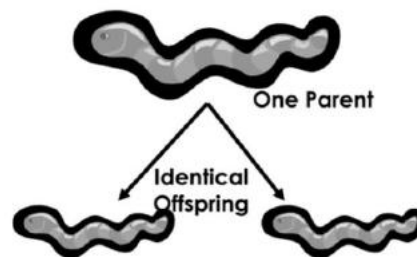
All living things must reproduce in order to produce offspring similar to themselves. This is essential for their survival. If they were unable to reproduce they would decrease in number and become extinct.

There are two types of reproduction, based on whether there is participation of one organism or two in the process of reproduction. When offspring is produced by a single parent with or without the involvement of gamete formation, the reproduction is asexual. When two parents (opposite sex) participate in the reproductive process and also involve fusion of male and female gametes, it is called sexual reproduction.

Also, there are two types of fertilization; internal (Insects, worms, reptiles, birds, mammals) and external (Amphibians, fish, jellyfish, clams, starfish, corals).

ASEXUAL REPRODUCTION

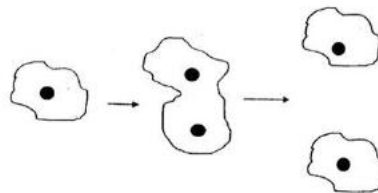
In this method, **a single individual** (parent) is capable of producing offspring. As a result, the offspring that are produced **are not only identical to one another but are also exact copies of their parent**. In asexual reproduction, **there is no cell fertilization**. Asexual reproduction involves only one parent. Young produced by asexual reproduction are identical to their parent and are usually referred to as **CLONE** because they are morphologically and genetically similar to their parents.



Asexual reproduction is common among single-celled organisms, and in plants and animals with relatively simple organizations. There are different types of asexual reproduction;

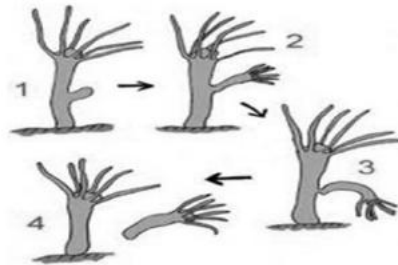
1. Binary fission

This is when **a cell divides into two halves** and each rapidly grows into an adult (e.g., *Amoeba*, *Paramecium*). The nucleus will first divide into two, followed by the cytoplasm (meaning) and then the cell wall/membrane



2. Budding

This is when **an offspring grows out of the body of the parent and then breaks off**. The **offspring organism is smaller than the parent**. The buds grow into fully matured individuals which eventually break away from the parent organism. Example: Hydra



3. Regeneration

If **a piece of a parent is detached, it can grow and develop into a completely new individual** through the process of cell division. Example: starfish, salamander

BENEFITS OF ASEQUAL REPRODUCTION

- ❖ There is no energy or time required to find a mate.
- ❖ The production of offspring is usually external so the parent can have LOTS of offspring in a very short amount of time.
- ❖ There is no parental care required; since the offspring are copies of the parent there is no “baby” stage
- ❖ Retention of useful characteristics/genes/traits
- ❖ Offspring establish faster/shorter life cycle

DISADVANTAGES OF ASEQUAL REPRODUCTION

- ❖ Lack of genetic variation. The beauty of nature is in diversity of genetic makeup
- ❖ Lowered resistance to disease
- ❖ Loss of hybrid potency. Hybrid usually have high resistance to disease
- ❖ Competition for resources due to overcrowding
- ❖ If a disease or change in environment occurs, all individuals will be affected. Inability to adapt to new environment

SEXUAL REPRODUCTION

Sexual reproduction involves formation of the male and female gametes. These gametes fuse to form the **zygote** which develops to form the new organism. Most **multicellular** organisms use **sexual reproduction** as a means of increasing their numbers. This involves **two** parents, male and female, each producing **sex cells** called **gametes**. Sexual reproduction increases the variety in a species.

In animals the male gametes are called **sperm** and are produced in the **testes**. The female gametes are called **eggs** and are produced in the **ovaries**. The fusion of male and female gametes to form a zygote is called fertilization. There are two types of fertilization; **External and internal**

In **external fertilization**, fusion of the male and female gametes takes place outside the body of the female e.g. amphibians and fish. In **internal fertilization**, union of gametes occurs inside the body of the female. Because of the fusion of male and female gametes, sexual reproduction results in offspring **that are not genetically identical to the parents or among themselves**.

ADVANTAGES OF EXTERNAL FERTILIZATION

- ❖ Large numbers of offspring produced per breeding season
- ❖ Female does not suffer gestation stress
- ❖ Mother does not need to care for the young except in a few species
- ❖ The surviving individuals are highly selected for better survival

DISADVANTAGES

- ❖ Many predators can feed on the eggs before or after fertilization
- ❖ **Fewer chances of fertilization/a lot of gametes wasted**
- ❖ The **development of the Embryo is at the mercy** of environment
- ❖ Large numbers of female gametes are required therefore female gets much exhausted

ADVANTAGES OF INTERNAL FERTILISATION

- ❖ More chances of fertilization
- ❖ Fewer predators of oval/fertilized eggs are protected in females body
- ❖ Stable internal environment
- ❖ Fewer gametes required

DISADVANTAGES

- ❖ Less number of offspring
- ❖ Less adapted for sudden change of environment after birth
- ❖ In mammals females suffer gestation stress

Topic 4

INTERRELATIONSHIPS OF ORGANISMS (COMPETITIONS, PARASITISM, PREDATION, SYMBIOSIS, COMMENSALISMS, MUTUALISM, SAPROPHYTISM).

Biological Interaction: This is the **relationship between two species in an ecosystem**. In addition, a biological interaction is the effect of two species living together in a community on each other in ecology. They might be of the **same species (intraspecific interactions)** or **different species (interspecific interactions)**. These impacts can be **short-term**, such as **pollination and predation**, or **long-term**, such as **extinction**, and both can have a significant impact on the evolution of the species involved. A **symbiosis** is a **long-term relationship**. **Mutualism**, which benefits both parties, to competition, which is destructive to both partners, are examples of symbioses.

There are five types of interactions between different species as listed below:

- 1) Competition & Predation
- 2) Commensalism
- 3) Parasitism
- 4) Mutualism
- 5) Saprophytism

Competition and Predation

Competition is when populations or even individuals compete for food resources. It is often referred to as exploitative or consumptive competition. **Competition** is most typically considered the interaction of individuals that vie for a common resource that is in limited supply, but more generally can be defined as the direct or indirect interaction of organisms that leads to a change in fitness when the organisms share the same resource. The outcome usually has negative effects on the weaker competitors. There are three major forms of competition. Two of them, **interference competition** and **exploitation competition**, are categorized as real competition. A third form, **apparent competition**, is not. Interference competition occurs directly between individuals, while exploitation competition and apparent competition occur indirectly between individuals.

When one entity hunts another animal to satisfy its nutritional requirements, it is referred to as predation.

Predation requires one individual, the predator, to kill and eat another individual, the prey. In most examples of this relationship, the predator and prey are both animals; however, protozoans are known to prey on bacteria and other protozoans and some plants are known to trap and digest insects (for example, pitcher plants). A predator is an entity that hunts its prey. For example, a snake eats a frog. Here snake is the predator and the frog is its prey. Typically, this interaction occurs between species (inter-specific); but when it occurs within a species (intra-specific) it is cannibalism. **Cannibalism** is quite common in both aquatic and terrestrial food webs. It often occurs when food resources are scarce, forcing organisms of the same species to feed on each other.

Commensalism

It is an imbalanced type of interaction wherein one individual benefits, while the other is neither harmed nor benefited. For example, orchids (examples of epiphytes) found in tropical rainforests grow on the branches of trees to access light, but the presence of the orchids does not affect the trees. Commensalism can be difficult to identify because the individual that benefits may have indirect effects on the other individual that are not readily noticeable or detectable. If the orchid from the previous example grew too large and broke off the branch or shaded the tree, then the relationship would become parasitic. There are four types of commensal associations:

- i. Inquilinism – An entity occupies the living habitat of another species (burrow, nest)
- ii. Chemical commensalism – A bacteria produces a chemical that nurtures another bacterium.
- iii. Phoresy – An organism tentatively attaches itself to another entity for transportation requirements.
- iv. Metabiosis – One entity is dependent on the other for survival.

Parasitism

Parasitism occurs when one individual, the parasite, benefits from another individual, the host, while harming the host in the process. Parasites feed on host tissue or fluids and can be found within (**endoparasites**) or outside (**ectoparasites**) of the host body. When the host is killed, this type of behaviour is referred to as parasitism. These parasites may be living on the surface of the host, often addressed as ectoparasites (fleas, leeches) while endoparasites live inside the host. Endoparasites can be subdivided into intracellular parasites (live inside cells) and intercellular parasites (live in spaces between cells). For example, different species of ticks are common ectoparasites on animals and humans. Parasitism is a good example of how species interactions are integrated. Parasites typically do not kill their hosts, but can significantly weaken them; indirectly causing the host to die via illness, effects on metabolism, lower overall health and increased predation potential. For instance, there is a trematode that parasitizes certain aquatic snails. Infected snails lose some of their characteristic behaviour and will remain on the tops of rocks in streams where food is inadequate and even during peaks of waterfowl activity, making them easy prey for the birds.

Mutualism

This is a symbiotic interaction where both or all individuals benefit from the relationship. Mutualism can be considered **obligate** or **facultative**. (Be aware that sometimes the term "symbiosis" is used specifically to mean mutualism.) Species involved in obligate mutualism cannot survive without the relationship, while facultative mutualistic species can survive individually when separated but often not as well. For example, leafcutter ants and certain fungi have **an obligate mutualistic relationship**. The **ant larvae eat only one kind of fungi**, and the **fungi cannot survive without the constant care of the ants**. As a result, the colonies' activities revolve around cultivating the fungi. They provide it with digested leaf material, can sense if a leaf species is harmful to the fungi, and keep it free from pests. A good example of a facultative mutualistic relationship is found between mycorrhizal fungi and plant roots. It has been suggested that 80% of vascular plants form relationships with mycorrhizal fungi. Yet the relationship can turn parasitic when the environment of the fungi is nutrient rich; because the plant no longer provides a benefit. Thus, the nature of the interactions between two species is often relative to the abiotic conditions and not always easily identified in nature

These interactions take place in three patterns:

- i. Facultative mutualism – Species survive on their own under favourable conditions
- ii. Obligate mutualism – One species is dependent for survival on the other
- iii. Diffusive mutualism – One entity can live with multiple partners

These relationships have three purposes: Defensive mutualism, Trophic mutualism, Dispersive mutualism

Topic 5

HABITATS

Habitat: It is the physical place where an organism normally/naturally lives. Specifically, it means the home where an organism (plant and animal) inhabit. The biologist who study habitats is called **ecologists**. Some organisms, Spearticularly migratory birds require more than one habitat. **Niche:** It is the functional role of an organism in a community, its job or position. Each species has a **potential niche** - what they could do with no competitors or resource limitations but due to competition and/or resource limitations, most organisms occupy a **realized niche**, the part of the fundamental niche that a species actually occupies in nature

TYPES OF HABITATS

There are basically tree types of habitat. They are terrestrial habitat, aquatic habitat and arboreal habitat.

1. Terrestrial habitat are exclusively land-based habitat. There are different types of terrestrial habitat distributed around various geological zones. They are as follows:

Forest habitat: A forest habitat consists of several plants, animals and microorganisms that live in coordination with the abiotic factors of the environment. Forests help in maintaining the temperature of the earth and are the major carbon sink.

Grassland habitat: In a grassland habitat, the vegetation is dominated by grasses and herbs. Temperate grasslands, savanna grasslands are some of the examples of grassland ecosystems.

Tundra habitat: Tundra habitat are devoid of trees and are found in cold climates or where rainfall is scarce. These are covered with snow for most of the year. The ecosystem in the Arctic or mountain tops is tundra type.

Desert habitat: Deserts are found throughout the world. These are regions with very little rainfall. The days are hot and the nights are cold.

2. Aquatic habitat: Aquatic habitats are habitat present in a body of water. These can be further divided into two types, namely: 1. Freshwater habitat 2. Marine habitat

Freshwater habitat: The freshwater habitat is an aquatic habitat that includes lakes, ponds, rivers, streams and wetlands. These have no salt content in contrast with the marine habitat.

Marine habitat: The marine habitat includes seas and oceans. These have a more substantial salt content and greater biodiversity in comparison to the freshwater habitat.

3. Arboreal Habitat: An arboreal habitat refers to a type of environment where organisms primarily live in trees. Many organisms spend most of their time in the tree, in the tree canopy. Arboreal habitats are also places in trees in which organisms are able to live and eat. These habitats include: roots of trees, canopies of rainforest, the branches of deciduous and coniferous trees, the leaves of trees, and even holes within the tree.

Topic 6

HEREDITY AND EVOLUTION

Introduction

Genetics is an important branch of biology as it is the science of heredity and variation. Heredity is the passing of traits from parents to offspring. It is a process through which an offspring acquires the traits (characteristics) of the parents. The hereditary units that are transmitted from one generation to the next are called **genes**. A gene contains coded information for the production of protein. The genetic information passed from one generation to another determines the structural, physiological, biochemical and behavioural traits of offspring. Chromosomes are considered as the physical basis of heredity because they have special organization and function. They contain DNA which is universally accepted as genetic or hereditary material.

Basis of Heredity

Heritable characteristics (traits) of organisms are determined by genes. Genes reside in the long molecules of deoxyribonucleic acid (DNA) of **chromosomes** of cells. Chromosomes are found in the nucleus of cells and they are made up of DNA and protein matrix. DNA, in conjunction with protein matrix forms **nucleoprotein** which organized into structures called chromosomes. A gene occupies a specific position on a chromosome called the gene **locus** (*plural; loci*). Each gene occurs in alternative form called **alleles** that are responsible for the expression of different forms of a trait. The alternative forms of genes determine the contrasting characteristics. The gene for pea colour, for example, has yellow and green alleles; the gene for pea shape has round and wrinkled alleles. Chromosomes occurs in pairs bearing alleles for a particular gene on the same loci. A pair of chromosomes bearing the same set of genes (alleles) is called **homologous chromosomes**.

The total number of chromosomes in the nucleus of a cell varies among organisms. In human, a total number 46 chromosomes (23 pairs of homologous chromosomes) are found in the nucleus of body cells. Out of these 23 pairs of chromosomes, 22 pairs are autosomal chromosomes and 1 pair is the sex chromosome (XX in female and XY in male). In sheep, there are 27 pairs of chromosomes, dog has 39 pairs, cow has 30 pairs, goat has 30 pairs, chicken 39 pairs, pig 19 pairs, cat 19 pairs, mouse 20 pairs, gorilla 24 pairs, elephant 28 pairs etc. However, the chromosomes of gametic cell (sperm cell and egg cell) do not occur in pairs. The human sperm cell and egg cell therefore contains 23 chromosomes (22 autosomal and 1 sex chromosome). Somatic cells have diploid number of chromosomes while gametic cells have haploid number of chromosomes. Other body cells apart from gametic cells, carry two copies of each gene. When the two copies of a gene are the same allele, the individual is homozygous for that gene (e.g. BB, RR, TT, tt, ss, rr, gg etc.). An individual is heterozygous for a gene if the two copies of alleles on homologous chromosomes at particular loci are different (e.g. Bb, Rr, Tt, Ss, and Gg etc.).

The **genotype** is a description of the allelic combination of the two copies of a gene present in an individual. It is the genetic constitution of an individual with respect to the alleles under consideration for example AA, Aa, aa. The **phenotype** is the observable form of the trait that an individual expresses. The

observable characteristics (phenotype) of an individual usually results from the interaction between the genotype and the environment in which development occurs.

The major scientific advance in the study of inheritance was made by an Austrian monk, **Gregor Mendel**, who discovered the basic principles of genetics. He was born in 1822, worked in Czech Republic and derived the basic laws of gene transmission.



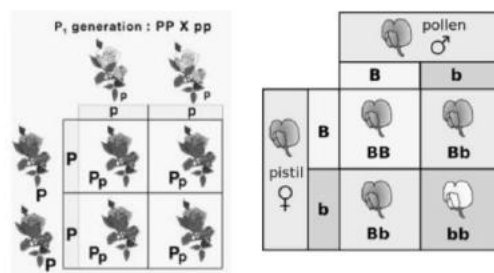
Gregor Mendel (1822-1884)

He developed his theory of heredity while working with garden peas (*Pisum sativum*). Gregor Mendel chose garden pea because of the following advantage: there were several varieties which had quite distinct characteristics; garden pea was easy to cultivate; the reproductive structure were completely enclosed and the plant was self-pollinating and artificial cross-breeding was possible with garden pea. He published his findings in 1866, just seven years after Charles Darwin publication.

Monohybrid inheritance and the law of segregation

Mendel carried out a series of matings between individual garden peas that differed in only one trait such as seed colour, stem length, seed shape etc. In each cross, one parent carries one form of the trait and the other parent carries an alternative form of the same trait. Mendel planted pure breeding green peas and pure breeding yellow peas and allow them to grow into the parental (P) generation. He later crossed stigma of green pea plants with the pollen from yellow-pea plants. He also performed the reciprocal cross using yellow pea plant stigma with green pea pollens. He observed the progeny peas of these crosses (original/initial and reciprocal) and found that in both cases, the peas were all yellow. These yellow peas, progenies of the parental (P) generation were the first filial (F₁) generation.

Mendel planted the F_1 yellow peas and allowed the F_1 plants to self-pollinate and grow into F_2 plants (second filial generation). Such experiments involving hybrids for a single traits are often called monohybrid crosses. The F_2 generation were in a ratio of 3 yellow: 1 green. The F_2 progenies resulting from the F_1 derived through reciprocal cross also produced a similar ratio of 3 yellow: 1 green. In Mendel's monohybrid crosses, one allele of gene was dominant and the other recessive. In the parent generation, one parent carried two dominant alleles while the other carried two recessive allele for the trait under consideration. Gametes are the specialized cells (egg and sperm cells) that carry genes between generations. During the formation of egg and sperm cells, the two copies of each gene in the parent separate (segregate) so that each gamete receives only one allele for each trait. At fertilization, male gamete with one allele unites at random with female gamete with the same or alternate allele, restoring the two copies of the gene for each trait in the fertilized egg or zygote.



One parent contains two copies of gene for each trait, however, only one of these two genes (alleles) is transmitted through a gamete to the offspring. For example, a plant that contains two allelic forms of a gene for seed shape (Rr), one for round (R) and one for wrinkled (r) seeds will transmit only one of these two alleles through gamete to its offspring.

	R	r
R	RR	Rr
r	Rr	rr

Law of Segregation - First law of Mendel

The two alleles for each trait separate (segregate) during gamete formation and only one copy of the alleles is present in the gamete. In other words, during meiosis, each pair of alleles separate (segregates) and each gamete receives one copy of all the alleles contained in the parental cell.

Dihybrid inheritance and the law of independent assortment

Mendel also made crosses with plants having two contrasting traits. He mated pure breeding plants grown from yellow round peas (YYRR) with pure breeding plants grown from green wrinkled peas (yyrr) and obtained a dihybrid F_1 generation (YyRr) showing only the two dominant phenotypes, yellow and round i.e. all the F_1 progenies are plants with yellow and round peas.

	Flower color	Flower position	Seed color	Seed shape	Pod shape	Pod color	Stem length
P	Purple	Axial	Yellow	Round	Inflated	Green	Tall
	White	Terminal	Green	Wrinkled	Constricted	Yellow	Dwarf
F_1	Purple	Axial	Yellow	Round	Inflated	Green	Tall

He then allowed these F_1 hybrids to self-fertilize to produce the F_2 generation. The F_2 generation produced was in the ratio of 9:3:3:1 (dihybrid ratio)

Apart from the yellow round and green wrinkled peas in the parental lines, Mendel also observed new phenotypic combination, yellow wrinkled and green round also emerged. The new phenotypic combination like these are called recombinant types. These combinations provided evidence that some shuffling of alleles had taken place. On the basis of these results, Mendel was able to state that the two pairs of characteristics (seed shape and colour) while combining in the F_1 generation separated and went into gamete formation independently. This forms the basis of Mendel's second law or principle.

Law of independent assortment Second law of Mendel

Any allele of a gene may combine with any allele of another gene during gamete formation. The segregation of alleles of one gene is independent of the alleles of other genes,

However, deviation to this Mendelian law of independent assortment was later discovered. Genes located closed to each other on a chromosome are said to be linked and therefore inherited together. Linked genes do not combine independently of each other. Linkage is an exception to the law of independent assortment.

EVOLUTION

The word 'evolution' means change over a period of time. Evolution is a change in the heritable characteristics of biological population over successive generation. It refers to observed changes in organism to their descent from a common ancestor and at a level to changes in gene frequency over time. The theory of evolution is one of the fundamental key stones of modern biology. The study of evolution provides a focus for investigations into the nature of life, origins of life and the great diversity of living things.

Theories of the origin of life on earth are:

Theory of special creation: This theory explained that life was created by a supernatural being at a particular time. This theory is supported by most of the world's major religion and civilizations and attribute the origin of life to a supernatural event at a particular time in the past.

Spontaneous generation: This theory was prevalent in ancient China, Babylon and Egypt. According to this theory, life arose from non-living thing matter on numerous occasion by a process of spontaneous generation. Aristotle (384-322BC) often called the founder of biology believed that life arose spontaneously. Aristotle's hypothesis of spontaneous generation assumed that certain particle of matter contained an active particle which could produce a living organism when condition were suitable. According to him, life arose from non- living matters on numerous occasions by a process of spontaneous generation.

Steady state theory: According to this theory, the earth has no origin. It has always been able to support life and has changed remarkably. This theory proposed that species never originated, they have always existed and number vary or can become extinct.

Cosmozoan theory: Life arrive on the earth planet from elsewhere.

Biochemical evolution: Life arose according to chemical and physical laws.

Paper 4 of 7 — Cyssa hub100LVL materials1st SemesterBIO 101Bio 101

Lecture/Notes · Google Drive harvest

Paper 5 of 7 — Cyssa hub100LVL materials1st SemesterBIO 101Cell Division - Definition Stages and Types B

Lecture/Notes · Google Drive harvest

Cell Division

By: **BD Editors** Reviewed by: **BD Editors** Last Updated: January 28, 2020

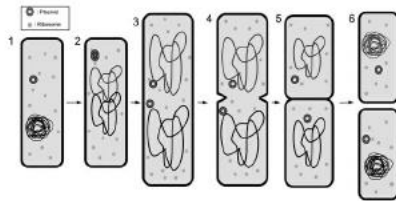
Cell Division Definition

Cell division is the process cells go through to divide. There are several types of cell division, depending upon what type of organism is dividing. Organisms have evolved over time to have different and more complex forms of cell division. Most prokaryotes, or bacteria, use binary fission to divide the cell. Eukaryotes of all sizes use mitosis to divide. Sexually-reproducing eukaryotes use a special form of cell division called meiosis to reduce the genetic content in the cell. This is necessary in sexual reproduction because each parent must give only half of the required genetic material, otherwise the offspring would have too much DNA, which can be a problem. These different types of cell division are discussed below.

Types of Cell Division

Prokaryotic Cell Division

Prokaryotes replicate through a type of cell division known as *binary fission*. Prokaryotes are simple organism, with only one membrane and no division internally. Thus, when a prokaryote divides, it simply replicates the DNA and splits in half. The process is a little more complicated than this, as DNA must first be unwound by special proteins. Although the DNA in prokaryotes usually exists in a ring, it can get quite tangled when it is being used by the cell. To copy the DNA efficiently, it must be stretched out. This also allows the two new rings of DNA created to be separated after they are produced. The two strands of DNA separate into two different sides of the prokaryote cell. The cell then gets longer, and divides in the middle. The process can be seen in the image below.

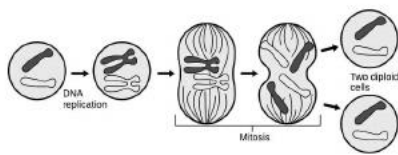


The DNA is the tangled line. The other components are labeled. Plasmids are small rings of DNA that also get copied during *binary fission* and can be picked up in the environment, from dead cells that break apart. These plasmids can then be further replicated. If a plasmid is beneficial, it will increase in a population. This is in part how antibiotic resistance in bacteria happens. The ribosomes are small protein structures that help produce proteins. They are also replicated so each cell can have enough to function.

Eukaryotic Cell Division: Mitosis

Eukaryotic organisms have membrane bound organelles and DNA that exists on chromosomes, which makes cell division harder. Eukaryotes must replicate their DNA, organelles, and cell mechanisms before dividing. Many of the organelles divide using a process that is essentially *binary fission*, leading scientist to believe that eukaryotes were formed by prokaryotes living inside of other prokaryotes.

After the DNA and organelles are replicated during interphase of the cell cycle, the eukaryote can begin the process of mitosis. The process begins during prophase, when the chromosomes condense. If mitosis proceeded without the chromosomes condensing, the DNA would become tangled and break. Eukaryotic DNA is associated with many proteins which can fold it into complex structures. As mitosis proceeds to metaphase the chromosomes are lined up in the middle of the cell. Each half of a chromosome, known as sister chromatids because they are replicated copies of each other, gets separated into each half of the cell as mitosis proceeds. At the end of mitosis, another process called cytokinesis divides the cell into two new daughter cells.



All eukaryotic organisms use mitosis to divide their cells. However, only single-celled organisms use mitosis as a form of reproduction. Most multicellular organisms are sexually reproducing and combine their DNA with that of another organism to reproduce. In these cases, organisms need a different method of cell division. Mitosis yields identical cells, but meiosis produces cells with half the genetic information of a regular cell, allowing two cells from different organisms of

the same species to combine.

Eukaryotic Cell Division: Meiosis

In sexually reproducing animals, it is usually necessary to reduce the genetic information before fertilization. Some plants can exist with too many copies of the genetic code, but in most organisms it is highly detrimental to have too many copies. Humans with even one extra copy of one chromosome can experience detrimental changes to their body. To counteract this, sexually reproducing organisms undergo a type of cell division known as *meiosis*. As before mitosis, the DNA and organelles are replicated. The process of meiosis contains two different cell divisions, which happen back-to-back. The first meiosis, *meiosis I*, separates homologous chromosomes. The homologous chromosomes present in a cell represent the two alleles of each gene an organism has. These alleles are recombined and separated, so the resulting daughter cells have only one allele for each gene, and no homologous pairs of chromosomes. The second division, *meiosis II*, separated the two copies of DNA, much like in mitosis. The end result of meiosis in one cell is 4 cells, each with only one copy of the genome, which is half the normal number.

Organisms typically package these cells into *gametes*, which can travel into the environment to find other gametes. When two gametes of the right type meet, one will fertilize the other and produce a zygote. The zygote is a single cell that will undergo mitosis to produce the millions of cells necessary for a large organism. Thus, most eukaryotes use both mitosis and meiosis, but at different stages of their lifecycle.

Cell Division Stages

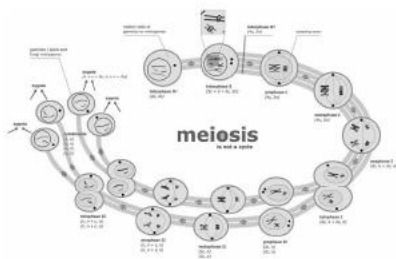
Depending upon which type of cell division an organism uses, the stages can be slightly different.

Mitosis Stages

Mitosis starts with *prophase* in which the chromosome is condensed. The cell proceeds to *metaphase* where the chromosomes are aligned on the metaphase plate. Then the chromosomes are separated in *anaphase* and the cell's *cytoplasm* is pinched apart during *telophase*. *Cytokinesis* is the final process that breaks the *cell membrane* and divides the cell into two.

Meiosis Stages

The stages of meiosis are similar to mitosis, but the chromosomes act differently. Meiosis has two phases, which include two separate cell divisions without the DNA replicating between them. *Meiosis I* and *meiosis II* have the same 4 stages as mitosis: prophase, metaphase, anaphase, and telophase. Cytokinesis concludes both rounds of meiosis.



In *prophase I*, the chromosomes are condensed. In *metaphase I*, the chromosomes line up across from their homologous pairs. When they are separated in *anaphase I* and

telophase I, there is only one form of each gene in each cell, known as a reduction division. Meiosis II proceeds in the same manner as mitosis, which sister chromatids dividing on the metaphase plate. By telophase II, there are 4 cells, each with half of the alleles as the parent cell and only a single copy of the genome. The cells can now become gametes and fuse together to create new organisms.

Quiz

1. Somatic cells are cells that fill the body, and must reproduce to repair damage. Gametic cells are cells that produce gametes. Which type of cell division do each type of cell undergo?

- A. Somatic= mitosis; Gametic= meiosis
- B. Somatic= mitosis; Gametic= meiosis and mitosis
- C. Somatic= mitosis and meiosis; Gametic= meiosis and mitosis

➤ Answer to Question #1

2. Mitochondria are organelles in cells that create ATP, a molecule used for energy. Mitochondria must replicate inside the cell, separate from mitosis or meiosis, to regulate the amount of energy being delivered. In mitochondria, there is a ring of DNA that controls the mitochondrial metabolism. This *mtDNA* is replicated, the mitochondria elongates, and divides in half. What type of cell division is this?

- A. Binary Fission
- B. Mitosis
- C. Meiosis

➤ Answer to Question #2

3. Evolution depends on the successful replication of DNA. In fact, all DNA on Earth comes from only one or two original cells, and most organisms are related to each other. What is responsible for the different forms

of life?

- A. Mutation
- B. Genetic Recombination
- C. Both

> Answer to Question #3



Cite This Article

MLA	Biologydictionary.net Editors. "Cell Division." <i>Biology Dictionary</i> , Biologydictionary.net, 15 Dec. 2016, https://biologydictionary.net/cell-division/.
APA	
Chicago	

Subscribe to Our Newsletter






Recognised MBA

Online MBA from the University of East London. Contact us today. UEL - Unicaf

Paper 6 of 7 — Cyssa hub100LVL materials1st SemesterBIO 101Nomenclature .pdf

Lecture/Notes · Google Drive harvest

Chapter 15

Nomenclature

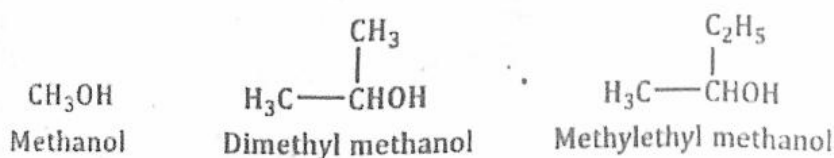
Before IUPAC International Union of pure and applied Chemistry nomenclature, organic compounds were named haphazardly. First naming to emerge was the trivial naming system, followed by rational, then semisystematic and last the systematic naming system.

15.1 Trivial Name

This system has no strict pattern. Compounds were named according to the origin, place of first discovery or even after the discoverer. Indole was prepared from a dye (indigotin) known to originate from India. The dye, a natural product was extracted and exported from that country. Grignard reagent was named after Grignard who first prepared the reagent and used it in synthesis. Some trivial names are still allowed by IUPAC especially in the naming of natural products. The systematic names of some compounds are so long and cumbersome that the names are better left at trivial level.

15.2 Rational Names

This naming system holds that organic substances were the products of the growing complexity of the simplest representative of a given homologous series. E.g. we are taking alcohols as example.



The radicals are enumerated in the order of methyl ethyl methanol. The nomenclature is still readily used today for very simple compounds.

15.3 Semi-systematic Names

This system combines the trivial and systematic system examples are the first four members of the hydrocarbon series methane, ethane, propane and butane. The ending indicates they are of the alkane but the number of carbon atoms cannot be derived from the name. We also have names such as cocaine, adrenaline, quinine, papaverine etc. Whose endings indicate the presence of nitrogen in the compound but the structures cannot be deduced from the names.

15.4 Systematic Names

This is the IUPAC naming system. It is scientific and commonly referred to as **substitutive naming** system. The names have three parts. The **prefix**, the **principal chain** and the **suffix**. The prefix is in numerals and radicals, the principal chain is the longest chain including the functional group. The suffix is the functional group which will be accompanied by a numeral when the functional group is not occupying position 1.

In 1892 several rules for this scientific nomenclature were approved at the international chemical congress in Geneva. It thus became known as 'Geneva nomenclature'. The rules of the nomenclature of open chain compounds were worked out in special great details.

In 1930 new rules for the naming of organic compounds were adopted in Liege (Belgium). They were known as 'Liege rules'. Work continued further bringing out the most complete rules of the commission on Nomenclature of organic compounds of the international union of pure and applied chemistry. (IUPAC).

15.5 IUPAC Rules

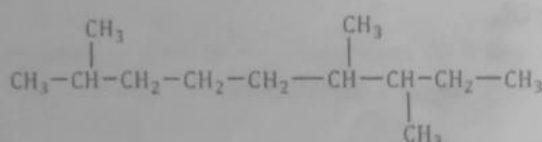
For Alkanes.

1. Locate the longest continuous carbon chain. This is the **principal chain**
2. Identify the radicals, these are groups that are not part of the principal chain. They are substituent's
3. Number the carbon atoms of the principal chain from left to right and right to left

4. The carbon atoms where there are radicals or substituents are **locants**. The locants are represented by numerals.
5. Arrange the locants and radicals at the **prefix**.
6. The principal chain and the suffix form the main name.
7. The numbering of the principal chain gives preference to the locant with the smaller first contact.
8. The prefix is arranged in alphabetical order, IUPAC also allows arrangement in order of complexity. The two arrangements must not be mixed.
9. The name ends in **-ane** for alkanes which is the suffix.
10. A few examples will suffice.

Care is taken to give the smallest possible number to substituents.

Let us try now and name the compound below:



2, 6, 7 - Trimethylnonane ✓ (1) Left to right

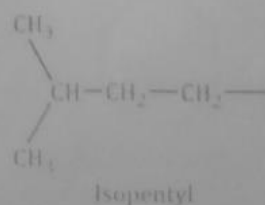
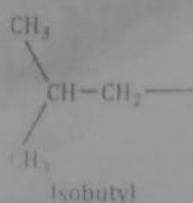
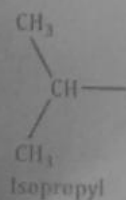
3, 4, 8 - Trimethylnonane X (2) Right to left

Addition of locants for both of them is 15. But the first locant contact is 2 for the first name (1) and for the second name (2) the first contact is 3. The correct name in obedience to rule 7 is (1).

15.6 The Radicals

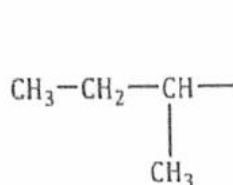
When the same radical appears two, three, four times or more, di, tri, tetra are attached to the radicals.

- The terms such as **iso-** is used for forked- end radicals.

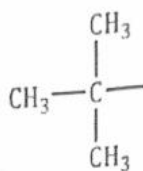


258

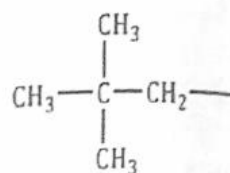
IUPAC also allows the terms *sec*-butyl, *tert* butyl and *neopentyl* to represent the following.



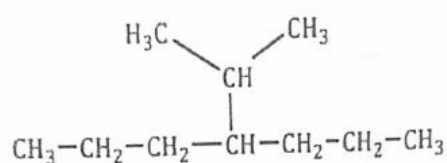
Secbutyl



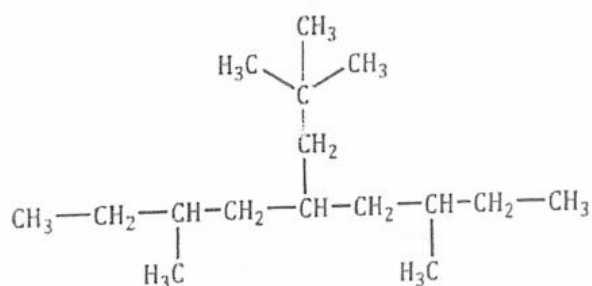
Tert butyl



Neopentyl



4-Isopropyl heptane



3,7-Dimethyl-5-neopentyl nonane

15.7 Cyclic Compounds

Cyclic compounds are accompanied by the prefix *cyclo*.



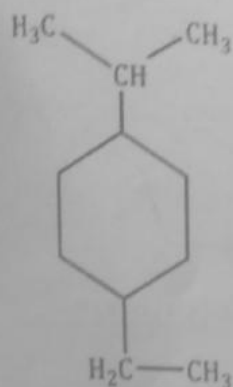
Cyclopentane



Cyclohexane



Cyclooctane

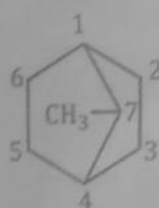


1-Ethyl-4-isopropylcyclohexane

15.8 Bicyclic Compounds

Numbering of the bicycle starts from the bridge head moves through the bigger side then inside the bridge. Care is taken to give substituents the smallest possible number.

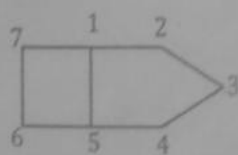
We shall take only three examples:



7-Methylbicyclo [2.2.1] heptane



Bicyclo [2.2.2] oct-2-ene

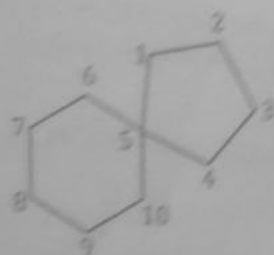


Bicyclo [3.2.0] heptane

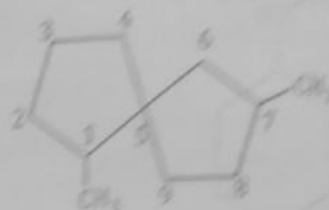
15.9 Spiro Compounds

The numbering of *spiro* compounds starts from the smaller cycle and moves round spirally to give the smallest number to substituents if any. The prefix is *spiro*.

260



Spiro [4.5] decane



1,7-Dimethyl spiro [4.4] nonane

15.10 Benzenes

The names of some benzenoids must be memorized, they are either trivial or semisystematic. Such compounds include:



Toluene



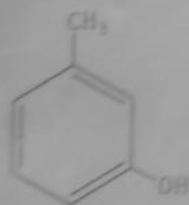
Phenol



Aniline



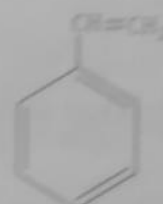
Anisole



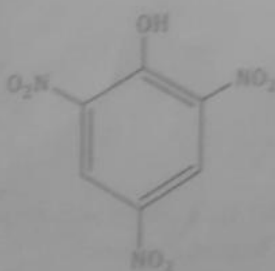
m-Cresol



Sulphanilic acid



Styrene
[vinylbenzene]



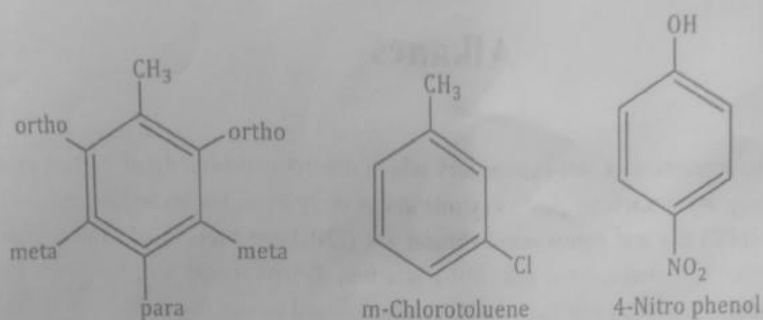
Picric acid



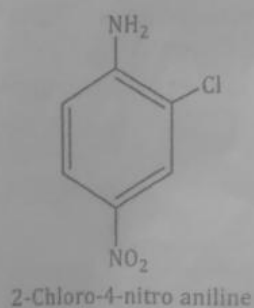
Acetophenone

When benzene has a substituent, the two carbon atoms closer on either side are the ortho (*o*-) positions. The next two positions on either side are meta (*m*-) positions and the position farthest away is the para (*p*-) position.

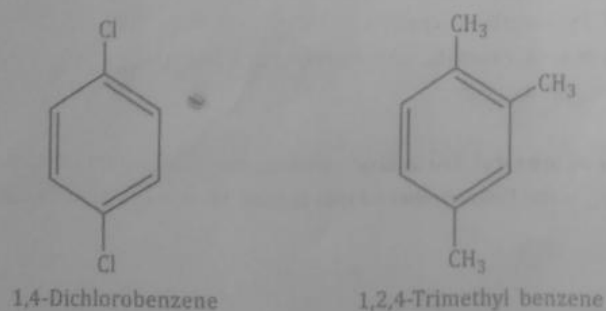
Numbering starts from the substituent and moves round clockwise or anticlockwise.



The use of numerals makes the naming of substituted benzenes easier.



When two or more similar substituents are on the benzene ring, the substituents form the prefix and benzene is the suffix.



Paper 7 of 7 — Cyssa hub100LVL materials1st SemesterBIO 101plant-cells.pdf

Lecture/Notes · Google Drive harvest

Plant Cells - Definition, Diagram, Structure & Function

The cell is the basic unit of life in all organisms. Like humans and animals, plants are also composed of several cells. The plant cell is surrounded by a cell wall which is involved in providing shape to the plant cell. Apart from the cell wall, there are other organelles that are associated with different cellular activities.

Let us have a detailed look at the plant cell, its structure, and functions of different plant cell organelles.

Plant Cell Definition

“Plant cells are eukaryotic cells with a true nucleus along with specialized structures called organelles that carry out certain specific functions.”

What is a Plant Cell?

Plant cells are eukaryotic cells that vary in several fundamental factors from other eukaryotic organisms. Both plant and animal cells contain nucleus along with similar organelles. One of the distinctive aspects of a plant cell is the presence of a cell wall outside the cell membrane.

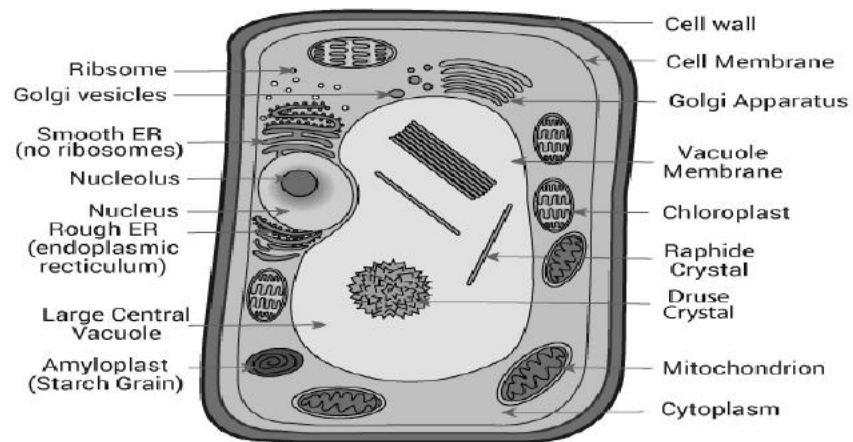
Read more: [Cells](#)

Plant Cell Diagram

The plant cell is rectangular and comparatively larger than the animal cell. Even though plant and animal cells are eukaryotic and share a few **cell organelles**, plant cells are quite distinct when compared to animal cell as they perform different functions. Some of these differences can be clearly understood when the cells are examined under an electron microscope.

Also Read: [Cellulose in Digestion](#)

Plant Cell



Plant Cell Diagram showing different cell organelles

Plant Cell Structure

Just like different organs within the body, plant cell structure includes various components known as cell organelles that perform different functions to sustain itself. These organelles include:

Cell Wall

It is a rigid layer which is composed of cellulose, glycoproteins, lignin, pectin, and hemicellulose. It is located outside the cell membrane. It comprises proteins, polysaccharides, and cellulose.

The primary function of the cell wall is to protect and provide structural support to the cell. The plant cell wall is also involved in protecting the cell against mechanical stress and to provide form and structure to the cell. It also filters the molecules passing in and out of the cell.

The formation of the cell wall is guided by microtubules. It consists of three layers, namely, primary, secondary and the middle lamella. The primary cell wall is formed by cellulose laid down by enzymes.

Also Read: Cell Wall

Cell membrane

It is the semi-permeable membrane that is present within the cell wall. It is composed of a thin layer of protein and fat.

The cell membrane plays an important role in regulating the entry and exit of specific substances within the cell.

For instance, cell membrane keeps toxins from entering inside, while nutrients and essential minerals are transported across.

Also Read: [Cell Wall and Cell Membrane](#)

Nucleus

The nucleus is a membrane-bound structure that is present only in eukaryotic cells. The vital function of a nucleus is to store DNA or hereditary information required for cell division, metabolism, and growth.

1. **Nucleolus:** It manufactures cell's protein-producing structures and ribosomes.
2. **Nucleopore:** Nuclear membrane is perforated with holes called nucleopore that allows proteins and nucleic acids to pass through.

Explore more: [The Nucleus](#)

Plastids

They are membrane-bound organelles that have their own DNA. They are necessary to store starch, to carry out the process of photosynthesis. It is also used in the synthesis of many molecules which form the building blocks of the cell. Some of the vital types of **plastids** and their functions are stated below:

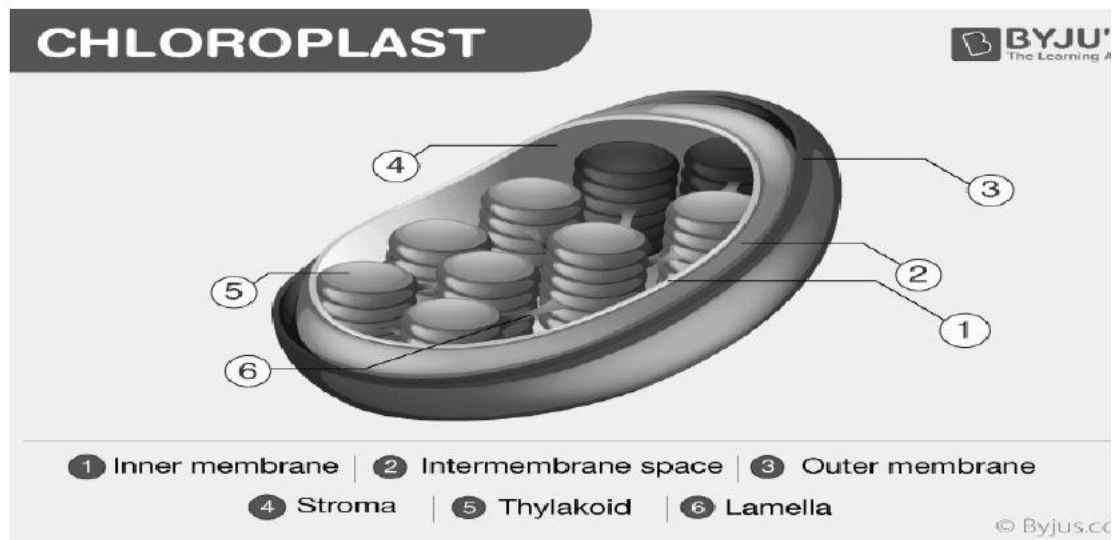
Leucoplasts

They are found in non-photosynthetic tissues of plants. They are used for the storage of protein, lipid, and starch.

Chloroplasts

It is an elongated organelle enclosed by phospholipid membrane. The chloroplast is shaped like a disc and the stroma is the fluid within the chloroplast that comprises a circular DNA. Each chloroplast contains a green coloured pigment called chlorophyll required for the process of photosynthesis. The chlorophyll absorbs light energy from the sun and uses it to transform carbon dioxide and water into glucose.

Also Read: [Chloroplasts](#)



Chromoplasts

They are heterogeneous, coloured plastid which is responsible for pigment synthesis and for storage in photosynthetic eukaryotic organisms. Chromoplasts have red, orange and yellow coloured pigments which provide colour to all ripe fruits and flowers.

Central Vacuole

It occupies around 30% of the cell's volume in a mature plant cell. Tonoplast is a membrane that surrounds central vacuole. The vital function of central vacuole apart from storage is to sustain turgid pressure against the cell wall. The central vacuole consists of cell sap. It is a mixture of salts, enzymes, and other substances.

Also read: Vacuoles

Golgi Apparatus

They are found in all eukaryotic cells which are involved in distributing synthesized macromolecules to various parts of the cell.

Explore more: Golgi Apparatus

Ribosomes

They are the smallest membrane-bound organelles which comprise RNA and protein. They are the sites for protein synthesis, hence, also referred to as the protein factories of the cell.

Explore more: Ribosomes

Mitochondria

They are the double-membraned organelles found in the cytoplasm of all eukaryotic cells. They provide energy by breaking down carbohydrate and sugar molecules, hence they are also referred to as the “Powerhouse of the cell.”

Explore more: [Mitochondria](#)

Lysosome

Lysosomes are called as suicidal bags as they hold digestive enzymes in an enclosed membrane. They perform the function of cellular waste disposal by digesting worn-out organelles, food particles and foreign bodies in the cell..

Also read: [Lysosomes](#)

Plant Cell Types

Cells of a matured and higher plant become specialized to perform certain vital functions that are essential for their survival. Few plant cells are involved in the transportation of nutrients and water, while others for storing food.

The specialized plant cells include parenchyma cells, sclerenchyma cells, collenchyma cells, xylem cells, and phloem cells.

Following are some of the different types of plant cells:

Collenchyma Cells

They are hard or rigid cells, which play a primary role in providing support to the plants when there is restraining growth in a plant due to lack of hardening agent in primary walls.

Sclerenchyma Cells

These cells are more rigid compared to collenchyma cells and this is because of the presence of a hardening agent. These cells are usually found in all plant roots and mainly involved in providing support to the plants.

Parenchyma Cells

Parenchyma cells play a significant role in all plants. They are the living cells of plants, which are involved in the production of leaves. They are also involved in the **exchange of gases**, production of food, storage of organic products and cell metabolism. These cells are typically more flexible than others because they are thinner.

Xylem Cells

Xylem cells are the transport cells in vascular plants. They help in the transport of water and minerals from the roots to the leaves and other parts of the plants.

Phloem Cells

Phloem cells are other transport cells in vascular plants. They transport food prepared by the leaves to different parts of the plants.

Refer more: [Plant Tissue System](#)

Plant Cell Functions

Plant cells are the building blocks of plants. Photosynthesis is the major function performed by plant cells.

Photosynthesis occurs in the chloroplasts of the plant cell. It is the process of preparing food by the plants, by utilizing sunlight, carbon dioxide and water. Energy is produced in the form of ATP in the process.

Few plant cells help in the transport of water and nutrients from the roots and leaves to different parts of the plants.

Also read: [Golgi Apparatus](#)

To more about what is a plant cell, its definition, structure, diagram, types and functions, keep visiting [BYJU'S Biology](#) website or download [BYJU'S](#) app for further reference.

Frequently Asked Questions

What is a plant cell?

A plant cell is a eukaryotic cell that contains a true nucleus and certain organelles to perform specific functions. However, some of the organelles present in plant cells are different from other eukaryotic cells.

What are the different types of plant cells?

The different types of plant cells include- collenchyma, sclerenchyma, parenchyma, xylem and phloem.

Which organelles are found only in plant cells?

The organelles found only in plant cells include- chloroplast, cell wall, plastids, and a large central vacuole. The chloroplasts contain a green pigment chlorophyll that is responsible for the process of photosynthesis.

What is the composition of a plant cell wall?

The cell wall of a plant is made up of cellulose. Cellulose is a long, linear polymer of several glucose molecules.

Where does photosynthesis occur in plant cells?

Photosynthesis occurs inside the chloroplast of the plant cells. Chloroplast consists of a green pigment called chlorophyll. The light reactions occur within the thylakoids of the chloroplast where the chlorophyll pigment is found.